

1. Where did you put each HITL type and why?

The validator node is between the end node and the generate answer node to validate the correct code and routing decision made. Then after the validation node where the human can act with the agentic pipeline, the trainer node logs each outcome with the human interaction as well. This way the trainer database with scale can be analyzed for patterns showing consistently incorrectly classifications to increase accuracy over time. The override node comes with the validator gate as the human can choose to override the decision made; or it can come after the end node where the has been made, but the decision has been made but can be overridden again.

2. What condition triggers the validator? Why that threshold?

The condition that triggers the validation gate is a critical severity or a confidence level of under 0.8. I chose a higher threshold for HITL because the LLM is typically overconfident and so 0.7 seems a bit low. This will result in more human intervention at first, however over time with the trainer node, ideally human intervention can decrease as the model gets better.

3. What information does the human see when reviewing? What's missing?

While reviewing the decision, the human can see the problem description, the issue classification code, confidence level, entity severity compared with real severity, and the routing decision being made. This human review only comes in for ambiguous and special cases. Additionally, If it's auto routed, the human can override it afterward.

4. If CBRE processes 10,000 calls/day — how does your design scale? What breaks first?

At this scale what breaks first is the validator gate. I think it depends on how many human validation tickets are being generated. With my own local testing, I would estimate around 20% of transcripts need a validator gate and so this would require an immensely large team.